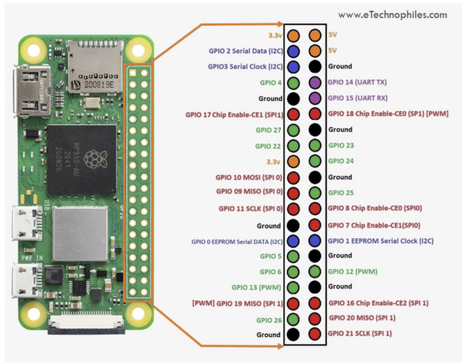
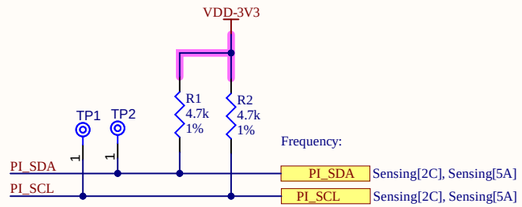
RPI Zero 2W GPIO pinout

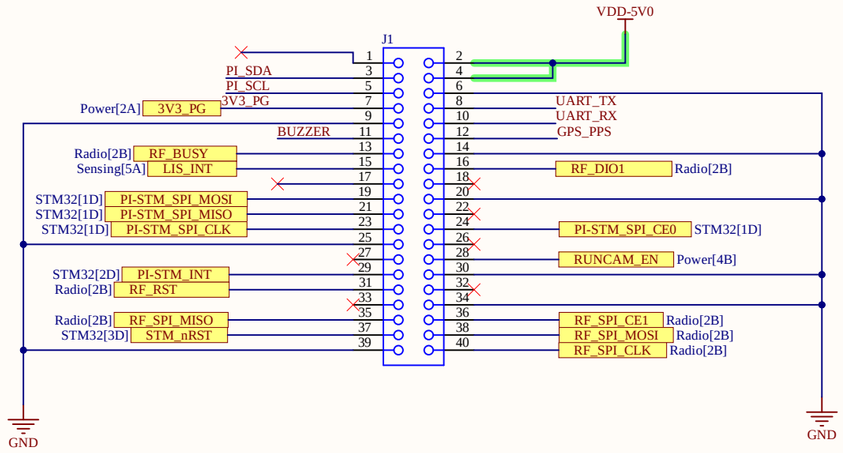


Raspberry Pi Zero 2W GPIO Pinout

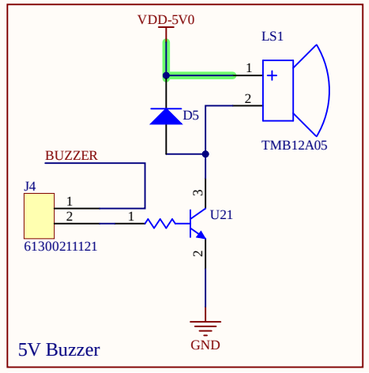
Pin Type	GPIO pins	Description
PWM pins	GPIO12, GPIO13, GPIO18, GPIO19	Used for Pulse Width Modulation
SPI pins	**SPI0:** GPIO9 (MISO), GPIO10 (MOSI), GPIO11 (SCLK), GPIO8(CE0)/ GPIO7(CE1) **SPI1:** GPIO19 (MISO), GPIO20 (MOSI), GPIO21 (SCLK), GPIO18 (CE0)/ GPIO17(CE1)/GPIO17(CE2)	Used for Serial Peripheral Interface
I2C Pins	**Data:** (GPIO2), **Clock:** (GPIO3) **EEPROM Data:** (GPIO0), **EEPROM Clock**: (GPIO1)	Used for Inter-Integrated Circuit communication
UART Pins	**TX**: (GPIO14) **RX**: (GPIO15)	To communicate between two microcontrollers or the computers.

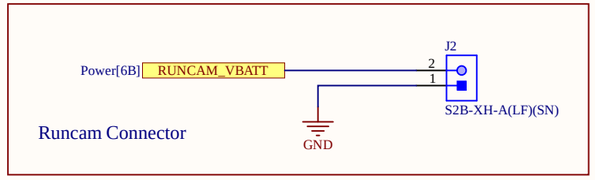


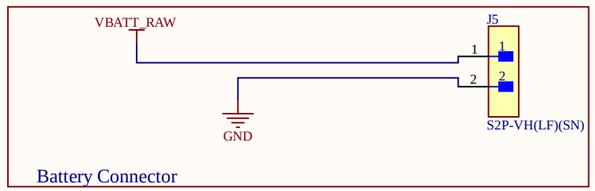
Raspberry Pi Zero 2W Schematic

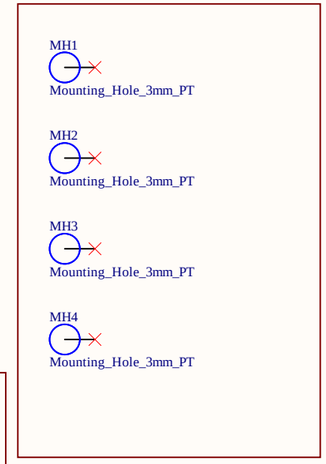


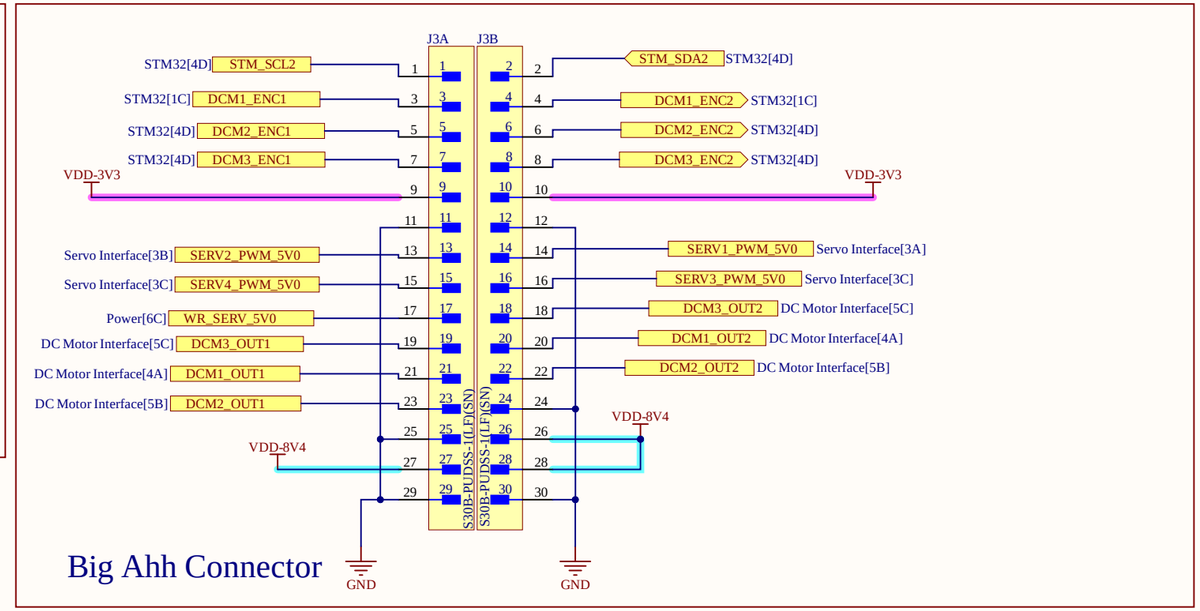
Raspbery Pi Zero 2W Interface

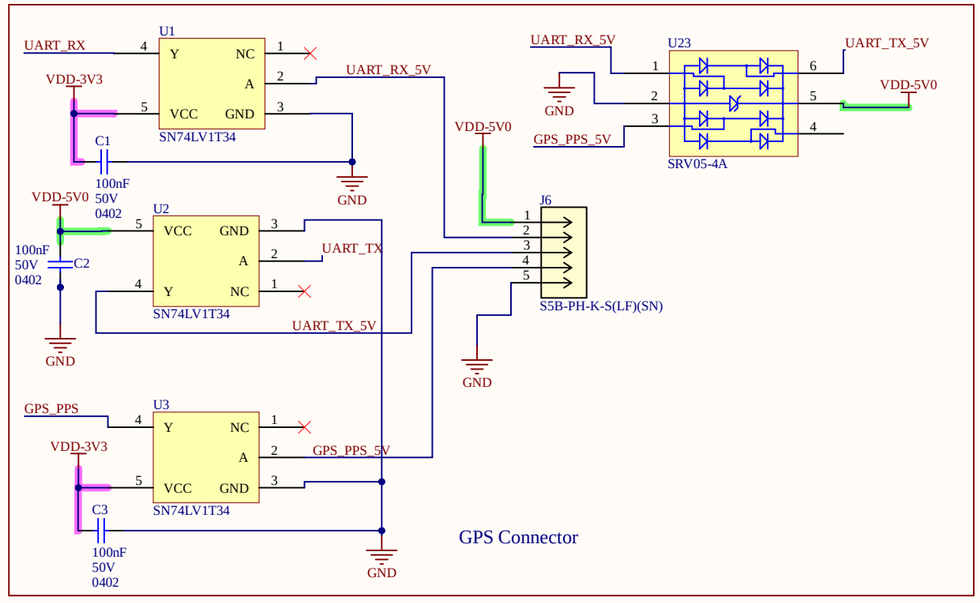










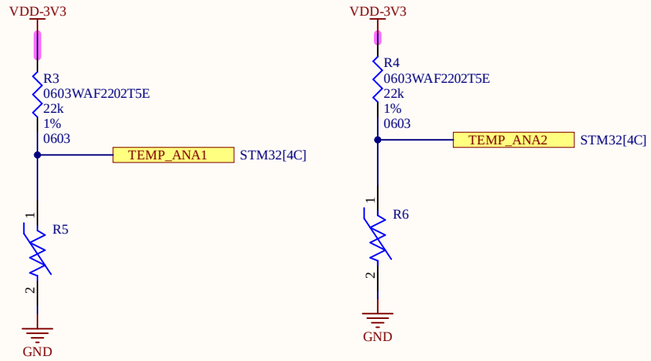


I2c Addr. Table

Bus	Component	Address
PI1	LIS2DH12TR	0011000b
PI1	INA260	1000100b
STM2	LIS3DHR Brk1	0011000b
STM2	LIS3DHR Brk2	0011001b
STM1	DRV8234	01100000b
STM2	DRV8235	01100001b
STM3	DRV8236	01100100b
STM4	DRV8237	01100101b
STM5	DRV8238	01101100b
STM6	DRV8239	01101101b

SHEET NAME: Interfaces.SchDoc	PROJECT: CRASHOUT.PrjPcb
ENGINEER: *	DATE: 4/15/2026

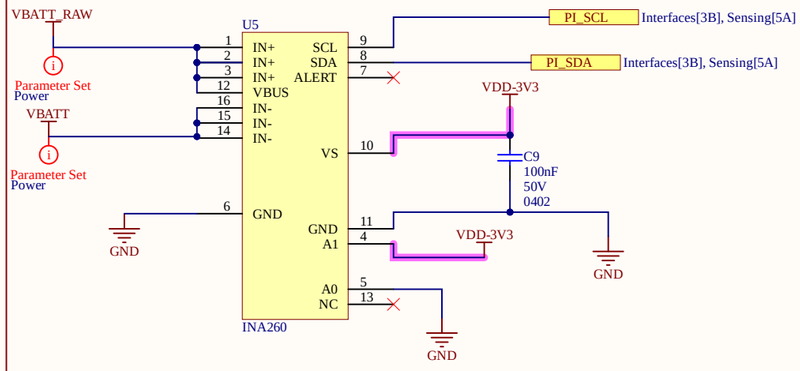
RIT LAUNCH INITIATIVE



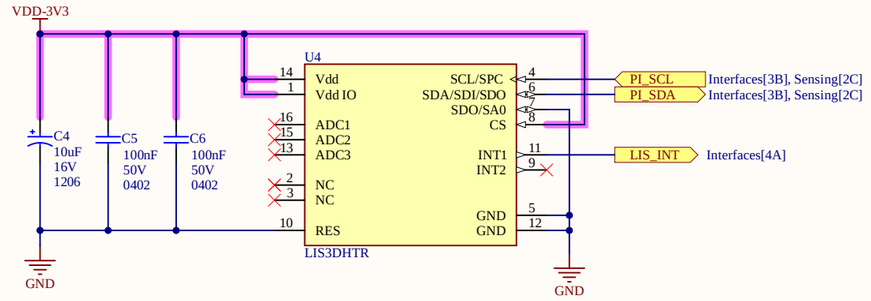
Temp Sense NTC Dividers

Battery Sense

i2c Address: 1000100 : 0x44



Accelerometer



i2c Address: 0011000b

The Slave ADdress (SAD) associated to the LIS3DH is 001100xb. The **SDO**/**SA0** pad can be used to modify the less significant bit of the device address. If the SA0 pad is connected to the voltage supply, LSb is '1' (address 0011001b) else if SA0 pad is connected to ground, the LSb value is '0' (address 0011000b). This solution permits to connect and address two different accelerometers to the same I²C lines.

2.3 Electrical characteristics

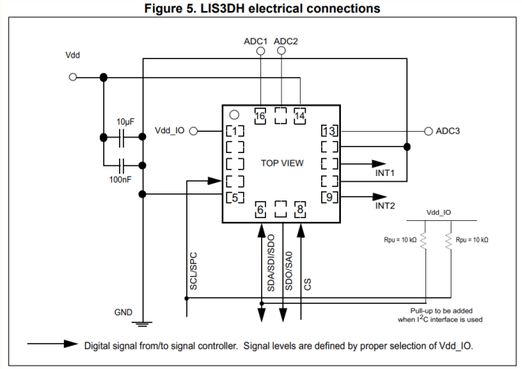
Vdd = 2.5 V, T = 25 °C unless otherwise noted [c]

Table 6. Electrical characteristics

Symbol	Parameter	Test conditions	Min.	Typ.[1]	Max.	Unit
Vdd	Supply voltage		1.71	2.5	3.6	V
Vdd_IO	I/O pins supply voltage[2]		1.71		Vdd+0.1	V

8	CS	SPI enable I²C/SPI mode selection: 1: SPI idle mode / I²C communication enabled 0: SPI communication mode / I²C disabled

Figure 5. LIS3DH electrical connections



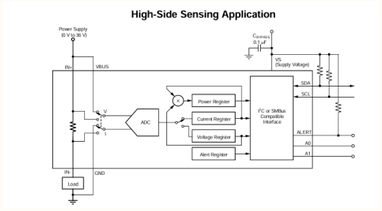
The device core is supplied through the Vdd line while the I/O pads are supplied through the Vdd_IO line. Power supply decoupling capacitors (100 nF ceramic, 10 µF aluminum) should be placed as near as possible to pin 14 of the device (common design practice).

All the voltage and ground supplies must be present at the same time to have proper behavior of the IC (refer to *Figure 5*). It is possible to remove Vdd maintaining Vdd_IO without blocking the communication bus, in this condition the measurement chain is powered off.

The functionality of the device and the measured acceleration data is selectable and accessible through the I²C or SPI interfaces.When using the I²C, CS must be tied high.

ADC1, ADC2 & ADC3 if not used can be left floating or connected to Vdd or GND.

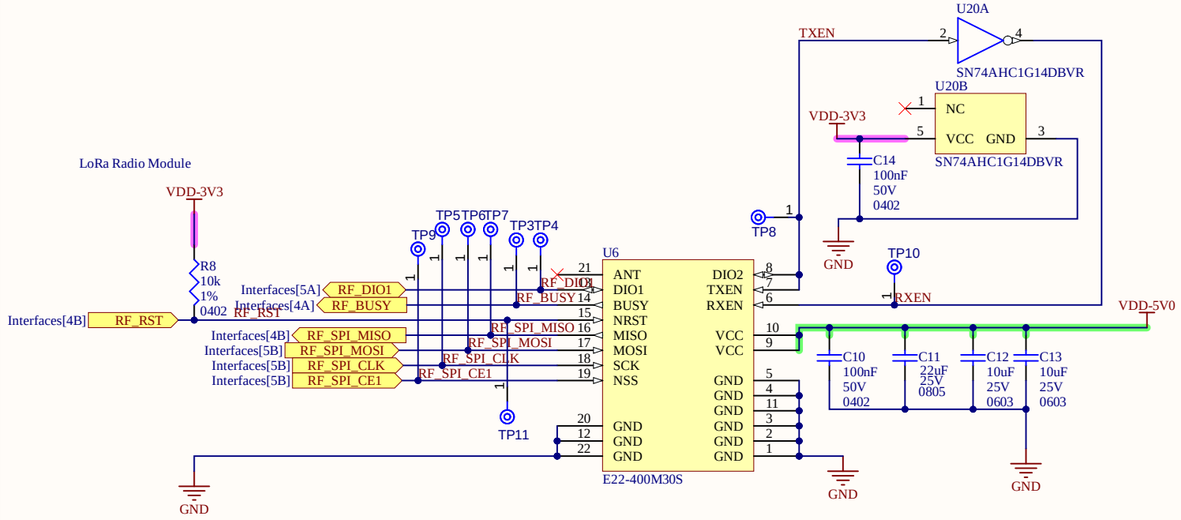
The functions, the threshold and the timing of the two interrupt pins (INT1 and INT2) can be completely programmed by the user through the I²C/SPI interface.



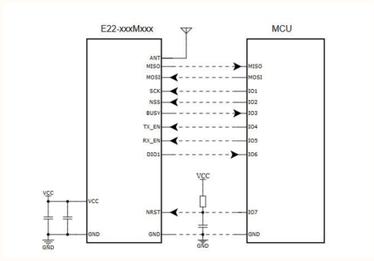
SHEET NAME: Sensing.SchDoc	PROJECT: CRASHOUT.PrjPcb
ENGINEER: *	DATE: 4/15/2026

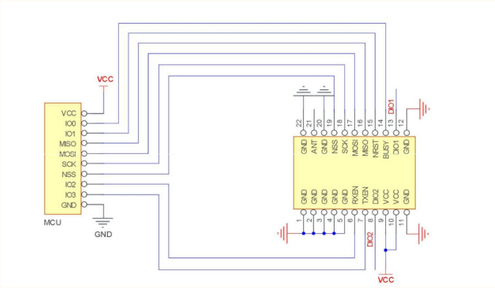
RIT LAUNCH INITIATIVE

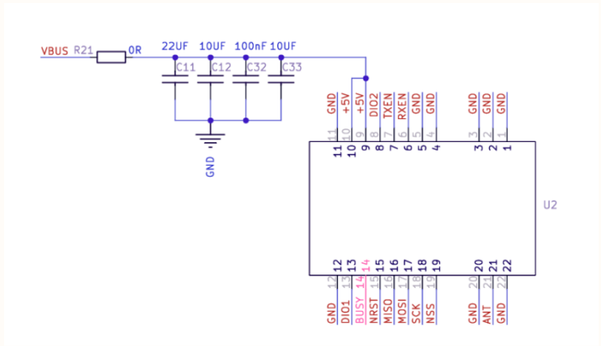
DIO2 has a double functionality. As DIO1, DIO2 can be used as a generic IRQ line and any IRQ can be routed through this pin. Also, DIO2 can be configured to drive an RF switch through the use of the command `SetDio2AsRfSwitchCtrl(...)`. In this mode, DIO2 is at a logical 1 during Tx and at a logical 0 in any other mode.



SN	Brief Connection between Module and Single Chip Microcomputer
1	The IO0、IO1、IO2、IO3 in MCU means the IO port of MCU.
2	DIO1、DIO2 is the normal IO port, which can be configured to multiple function. The DIO2 can connect with TXEN， but not with the IO port of MCU. It's used to control RF switch, see more details in SX1262 datasheet. Also It can be floated when not in use.
3	The DIO3 is powered by a 32MHz TCXO crystal.
4	Ensure the good grounding, large area of paving, small the power supply ripple, the filter capacitor should be increased and as close as possible to the VCC and GND pins of module.
5	It is recommended to add 200R protection resistor to RXD/TXD of external MCU.

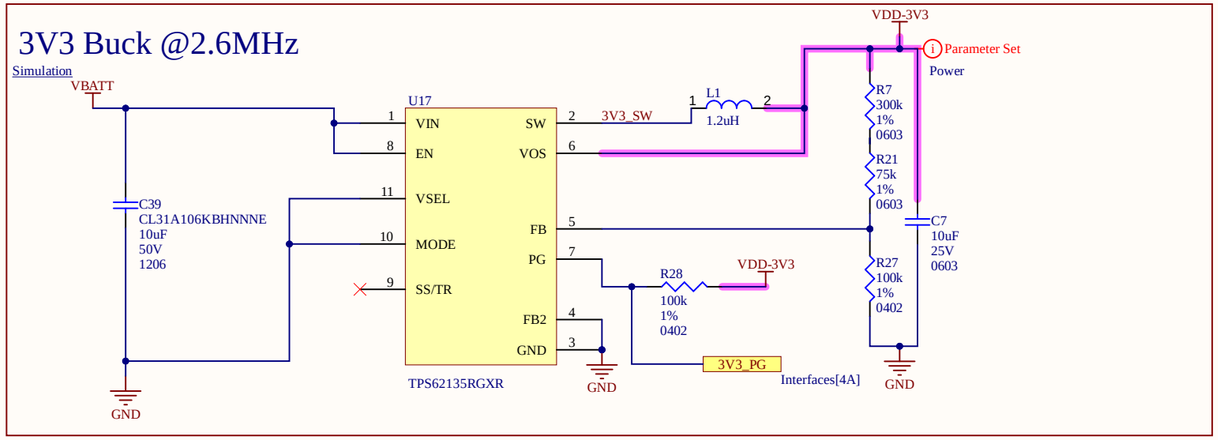






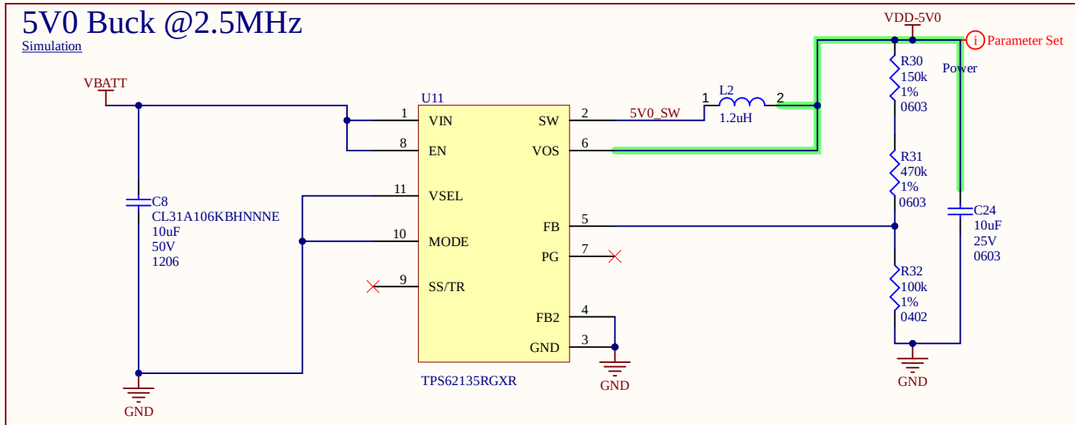
3V3 Buck @2.6MHz

Simulation



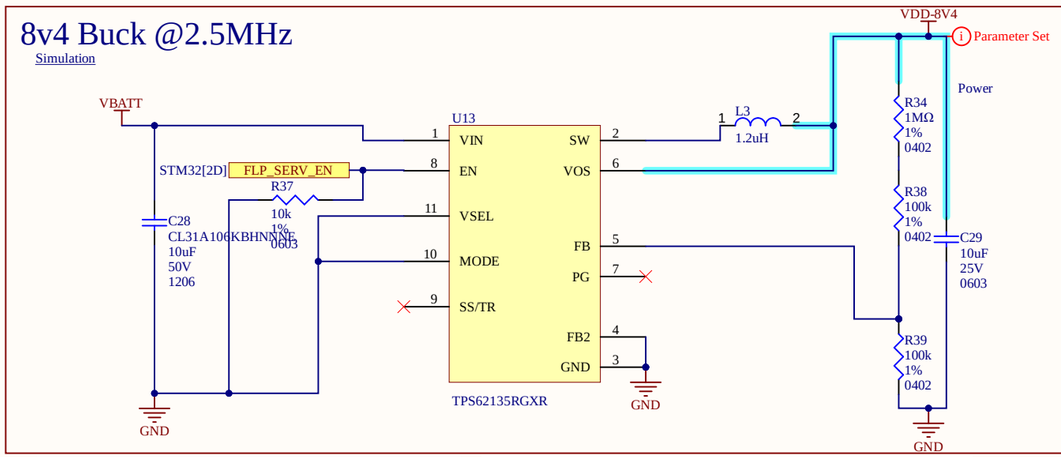
5V0 Buck @2.5MHz

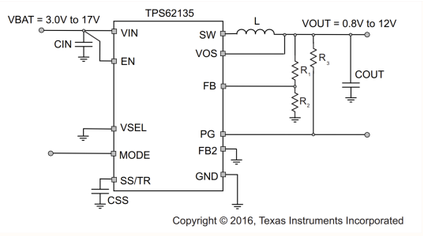
Simulation



8v4 Buck @2.5MHz

Simulation





12.2 Layout Example

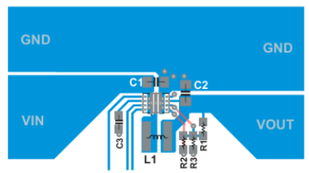
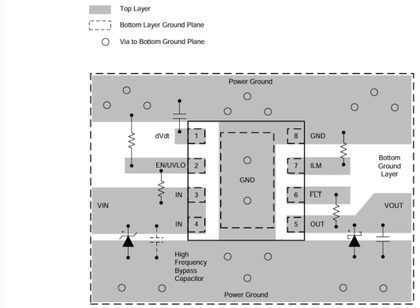


Figure 92. Layout

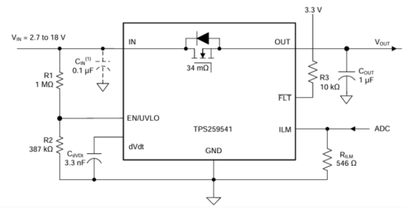
11.2 Layout Example



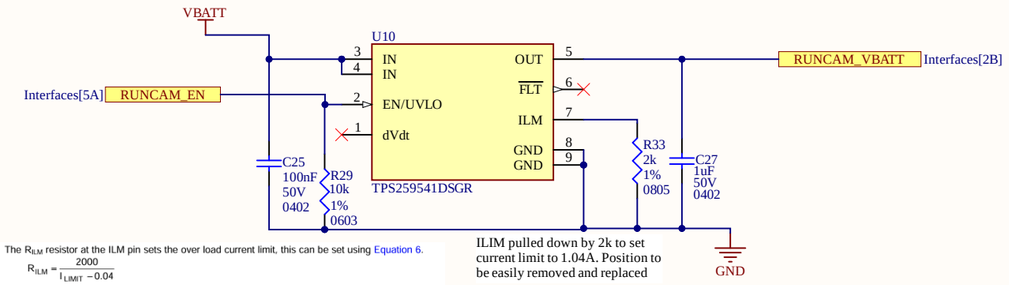
(1) Optional: Needed only to suppress the transients caused by inductive load switching

Figure 68. TPS2595xx Layout

9.2 Typical Application



Runcam E-Fuse

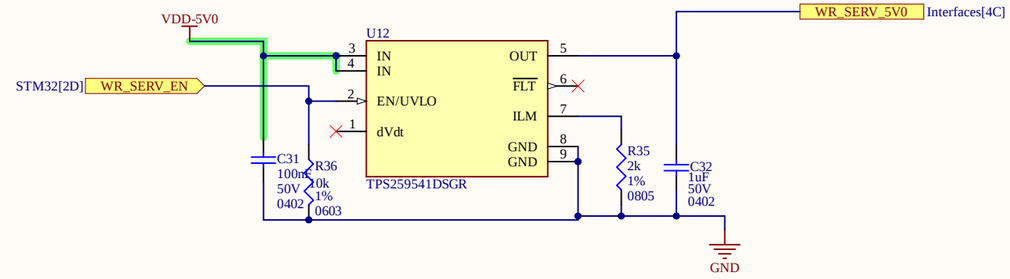


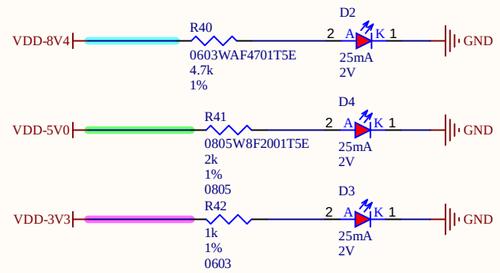
The R_{ILM} resistor at the ILM pin sets the over load current limit, this can be set using Equation 6.

$$R_{ILM} = \frac{2000}{I_{LIMIT} - 0.04}$$

ILIM pulled down by 2k to set current limit to 1.04A. Position to be easily removed and replaced

Wrist Servo E-Fuse





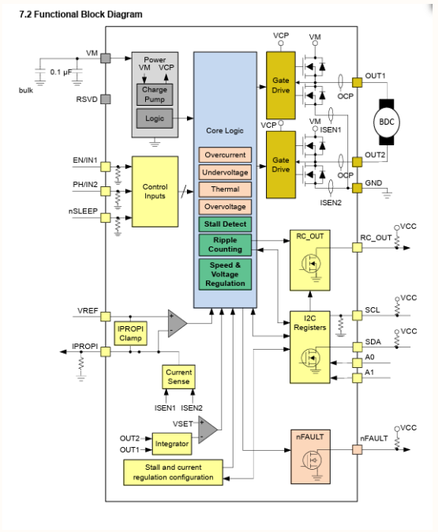
SHEET NAME: Power.SchDoc	PROJECT: CRASHOUT.PrjPcb
ENGINEER: *	DATE: 4/15/2026

RIT LAUNCH INITIATIVE

Table 7-1. Recommended External Components

COMPONENT	PIN 1	PIN 2	RECOMMENDED
C_{VM1}	VM	GND	0.1-µF, low ESR ceramic capacitor, VM-rated.
C_{VM2}	VM	GND	Bulk Capacitance, VM-rated.
R_{IPROPI}	IPROPI	GND	Resistor from IPROPI pin to GND, sets the current regulation level.
R_{nFAULT}	System VCC	nFAULT	10 kΩ
R_{RC_OUT}	System VCC	RC_OUT	10 kΩ
$R_{Pull\text{-}up}$	SDA, SCL, A0, A1	VM	2.2 kΩ

7.2 Functional Block Diagram



$$C_{BULK} > k \times \Delta I_{MOTOR} \times T_{PWM} / \Delta V_{SUPPLY} \quad (6)$$

Where:

C_{BULK} is the bulk capacitance

k is a scale factor to account for the ESR for typical capacitors in this type of application; based on the lab measurements above with DRV8718 EVM, k of about 3 is practical for these cases.

ΔI_{MOTOR} is the expected variation in motor current, $i_{max} - i_{min}$

T_{PWM} is the PWM period which is the reciprocal of the PWM frequency

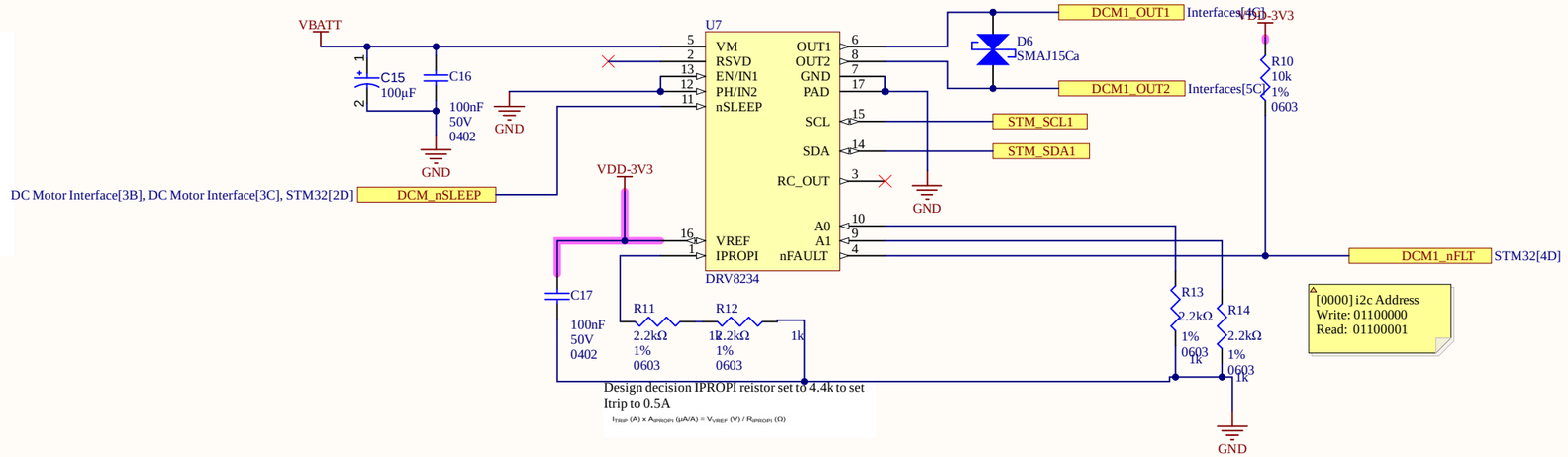
ΔV_{SUPPLY} is the allowable variation in the motor supply voltage

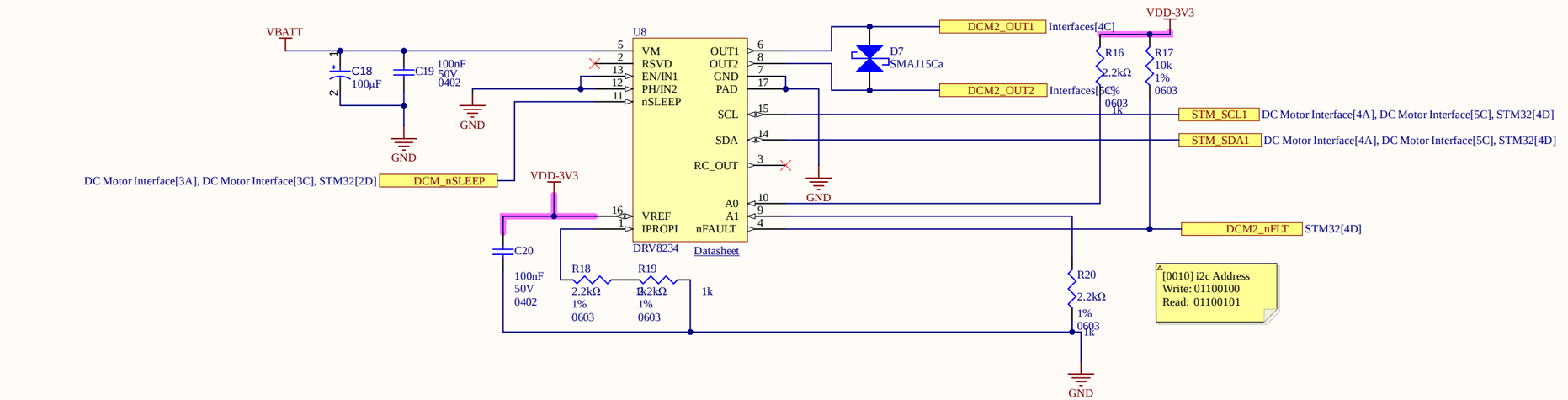
Figure 3-7 plots several data points and applies this general guideline, showing relatively good agreement.

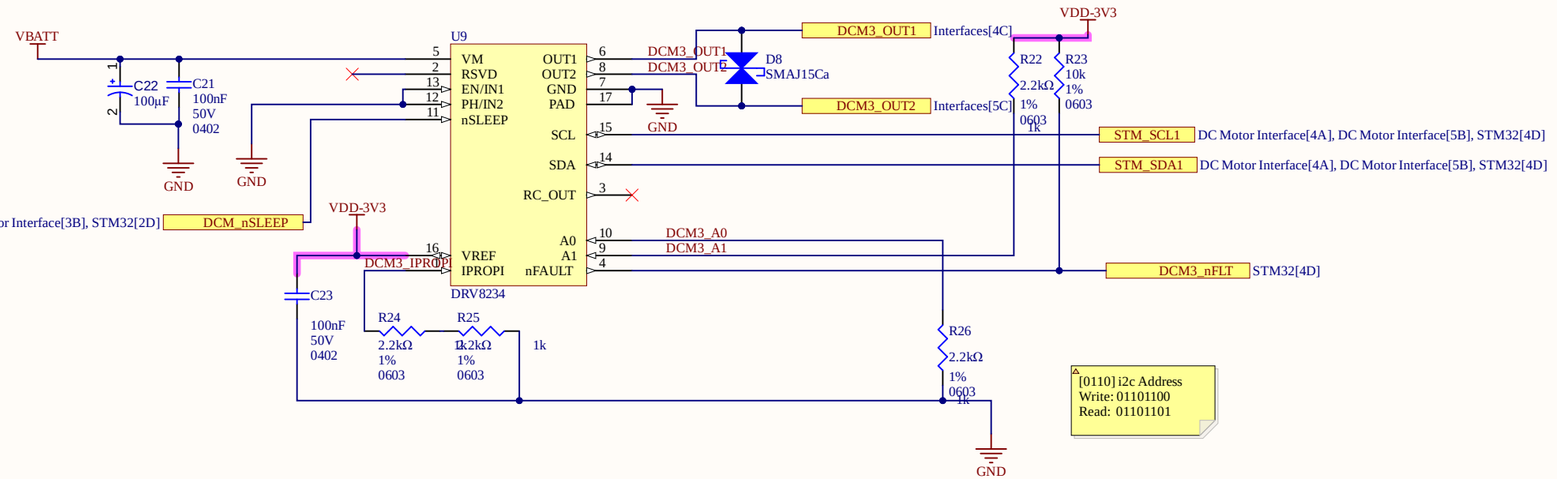
Experienced engineers often use general guidelines about bulk capacitance to select the capacitor values. One such guideline says to use at least 1 to 4µF of capacitance for each Watt of motor power. For example, a motor which draws 10 Amps from a 12V supply has a power of 120 Watts, leading to bulk capacitance of 120 to 480µF, using this general guideline. We dig a little deeper and see what further discussion supports those estimates.

Table 7-24. Device Addresses

A1 Pin	A0 Pin	A3A2A1A0 bits	ADDRESS (WRITE)	ADDRESS (READ)
0	0	0000b	0x60h	0x61h
0	High-Z	0001b	0x62h	0x63h
0	1	0010b	0x64h	0x65h
High-Z	0	0011b	0x66h	0x67h
High-Z	High-Z	0100b	0x68h	0x69h
High-Z	1	0101b	0x6Ah	0x6Bh
1	0	0110b	0x6Ch	0x6Dh
1	High-Z	0111b	0x6Eh	0x6Fh
1	1	1000b	0x70h	0x71h

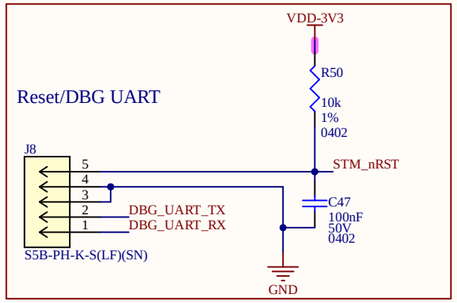


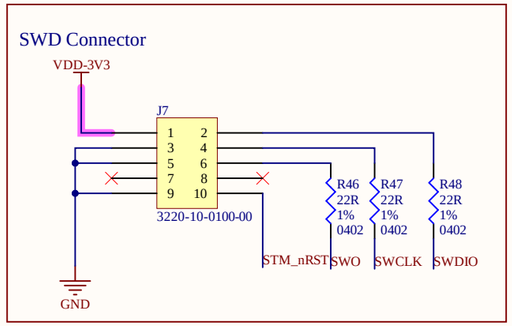


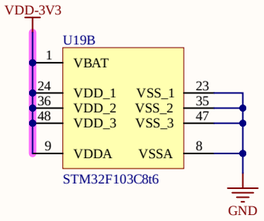


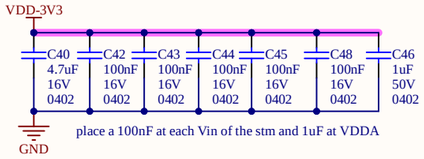
SHEET NAME: DC Motor Interface.SchDoc	PROJECT: CRASHOUT.PrjPcb
ENGINEER: *	DATE: 4/15/2026

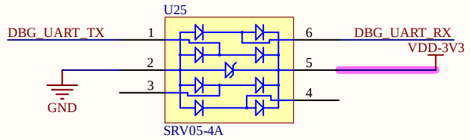
RIT LAUNCH INITIATIVE

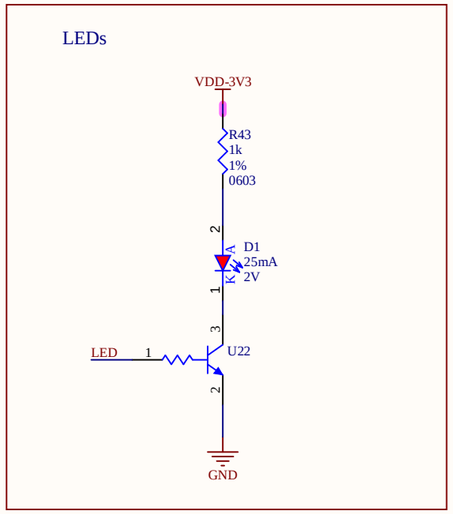


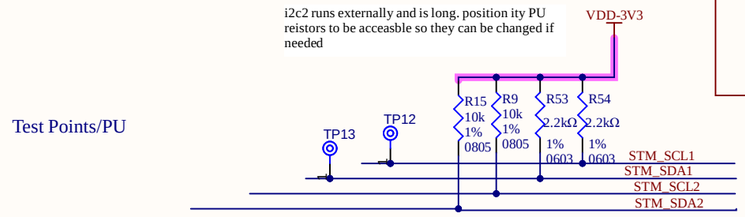


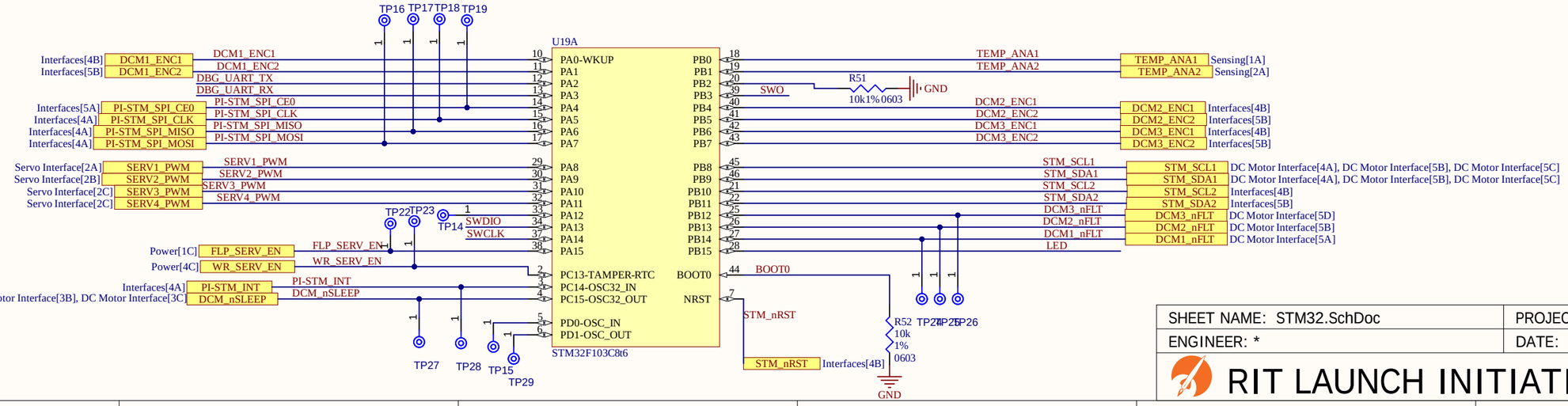






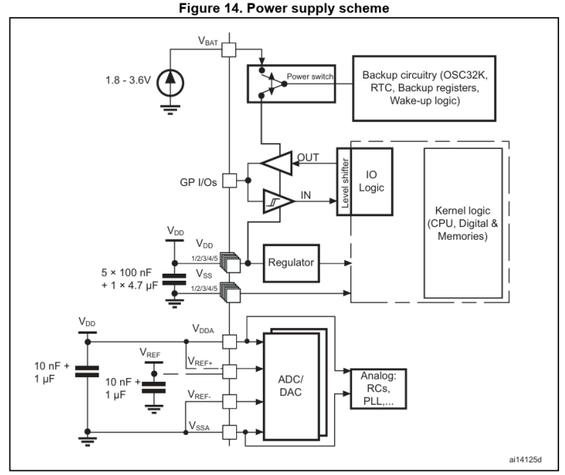






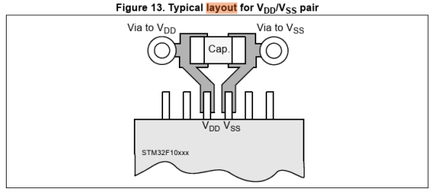
5.1.6 Power supply scheme

Figure 14. Power supply scheme



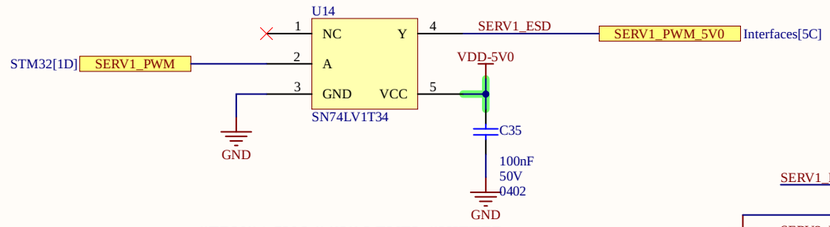
Caution: In *Figure 14*, the 4.7 µF capacitor must be connected to V_{DD3}.

Figure 13. Typical layout for V_{DD}/V_{SS} pair

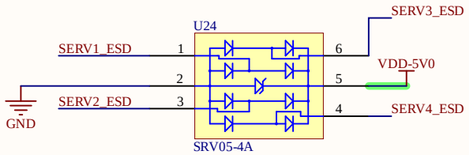


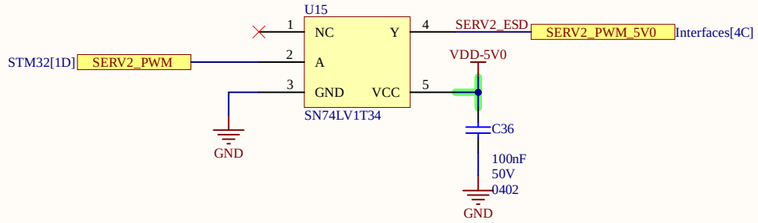
SHEET NAME: STM32.SchDoc	PROJECT: CRASHOUT.PrjPcb
ENGINEER: *	DATE: 4/15/2026

RIT LAUNCH INITIATIVE

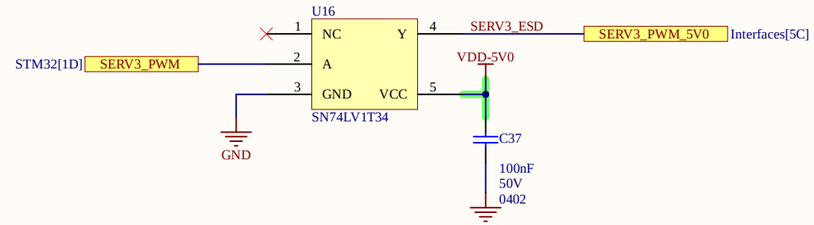


SERVO1 5V LOGIC LEVEL SHIFTER

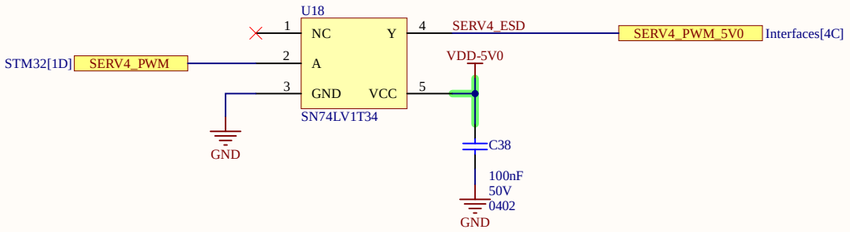




SERVO2 5V LOGIC LEVEL SHIFTER



SERVO3 5V LOGIC LEVEL SHIFTER



SERVO4 (wrist) 5V LOGIC LEVEL SHIFTER

SHEET NAME: Servo Interface.SchDoc	PROJECT: CRASHOUT.PrjPcb
ENGINEER: *	DATE: 4/15/2026

RIT LAUNCH INITIATIVE